# B. F1 Champions

time limit per test: 2 seconds
memory limit per test: 256 megabytes

Formula One championship consists of series of races called Grand Prix. After every race drivers receive points according to their final position. Only the top 10 drivers receive points in the following order 25, 18, 15, 12, 10, 8, 6, 4, 2, 1. At the conclusion of the championship the driver with most points is the champion. If there is a tie, champion is the one with most wins (i.e. first places). If a tie still exists, it is chosen the one with most second places, and so on, until there are no more place to use for compare.

Last year another scoring system was proposed but rejected. In it the champion is the one with most wins. If there is tie, champion is the one with most points. If a tie still exists it is proceeded the same way as in the original scoring system, that is comparing number of second, third, forth, and so on, places.

You are given the result of all races during the season and you are to determine the champion according to both scoring systems. It is guaranteed, that both systems will produce unique champion.

## Input

The first line contain integer $t$ $(1 \le t \le 20)$, where $t$ is the number of races. After that all races are described one by one. Every race description start with an integer $n$ $(1 \le n \le 50)$ on a line of itself, where $n$ is the number of clasified drivers in the given race. After that $n$ lines follow with the classification for the race, each containing the name of a driver. The names of drivers are given in order from the first to the last place. The name of the driver consists of lowercase and uppercase English letters and has length at most 50 characters. Comparing of names should be case-sensetive.

## Output

Your output should contain exactly two line. On the first line is the name of the champion according to the original rule, and on the second line the name of the champion according to the alternative rule.

## Examples

**input** Copy

```
3
3
Hamilton
Vettel
Webber
2
Webber
Vettel
2
Hamilton
Vettel
```

**output** Copy

```
Vettel
Hamilton
```

**input** Copy

```
2
7
Prost
Surtees
Nakajima
Schumacher
Button
DeLaRosa
Buemi
```

→ **Attention**

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

**Codeforces Beta Round 24**

**Finished**

Practice

→ **Virtual participation**

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ **Clone Contest to Mashup**

You can clone this contest to a mashup.

Clone Contest

→ **Submit?**

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ **Last submissions**

| Submission | Time | Verdict |
|---|---|---|
| 372130722 | Apr/22/2026 15:09 | Accepted |
| 372130555 | Apr/22/2026 15:08 | Accepted |
| 372130319 | Apr/22/2026 15:05 | Accepted |

→ **Problem tags**

```
8
Alonso
Prost
NinoFarina
JimClark
DeLaRosa
Nakajima
Patrese
Surtees
```

**output** Copy

```
Prost
Prost
```

### Note

It is not guaranteed that the same drivers participate in all races. For the championship consider every driver that has participated in at least one race. The total number of drivers during the whole season is not more then $50$.

implementation *1500

No tag edit access

### → Contest materials

- Announcement
- Tutorial

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